

Notes: Ahern (JFE, 2017)

Information Networks: Evidence from Illegal Insider Trading Tips

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1 Big picture

- **Question.** How does valuable private information spread among investors, and how do social networks affect who receives information, how prices move, and who earns trading gains?
- **Setting.** The paper studies illegal insider trading tips because SEC and DOJ case documents provide rare direct observations of person-to-person communication among investors.
- **Data contribution.** Ahern hand-collects data from insider-trading cases filed by the SEC and DOJ, producing a data set of 183 insider-trading networks, 622 people, 1,139 tips, 465 corporate events, and about \$928 million in illegal profits.
- **Core fact.** Inside information flows through strong social ties. Family, friends, business associates, and geographic proximity are central; the median tipper-tippee distance is only about 26 miles.
- **Diffusion pattern.** Tips usually start with corporate executives or other original sources and move outward through the network. Buy-side analysts and portfolio managers become dominant only after several links.
- **Hierarchy pattern.** Information tends to flow from lower-status people to higher-status people: subordinates to bosses, younger tippers to older tippees, and children to parents.
- **Market result.** Insider trading moves prices toward fundamental values before public announcements. More insider trading on a given day predicts stronger event-stock returns.
- **Gains result.** More central inside traders earn higher returns and profits. Receiving more tips is more valuable than merely giving more tips.
- **Network structure.** Large insider networks are sparse and branch-like rather than compact hubs. This looks similar to criminal networks, where trust and secrecy shape communication.
- **Main takeaway.** Social networks are not just background demographics. They are the mechanism through which high-quality financial information is transmitted, priced, and monetized.

2 Data and measurement

2.1 Why illegal insider trading is useful for studying information networks

- Most empirical work on investor social networks uses proxies such as geography, schools, employment, or correlated trades.
- Those proxies can confuse information transmission with homophily: similar people may behave similarly without talking to each other.
- Illegal insider trading cases reveal actual communications between named tippers and tippees.
- Because the law requires the information to be material and nonpublic, the sample focuses on valuable information rather than noise or public media reports.

Interpretation. The setting is unusual because it observes the communication graph directly. That makes the paper a rare direct test of information diffusion among investors.

2.2 SEC and DOJ case data

- The paper starts from all insider-trading cases filed by the SEC and DOJ from 2009 to 2013.
- The underlying corporate events span 1996–2013.
- Case documents include civil complaints, criminal complaints, DOJ information documents, PACER filings, and investigative-source documents.
- The author reads and codes the documents by hand because the data are narrative rather than tabular.
- The case documents identify names, locations, employers, ages, relationships, original sources, timing of communication, securities traded, trade dates, amounts invested, and realized gains.

Interpretation. The paper converts legal narratives into a person-level and link-level network data set. The unit of observation can be a person, a tipper-tippee link, an event, a network, or a trading day.

2.3 LexisNexis Public Records Database

- The LexisNexis Public Records Database (LNPRD) provides demographic information, addresses, property ownership, employers, family links, and associates.
- It is used to identify people named in legal documents and to construct counterfactual social networks.
- For each tipper, the author observes family members and associates who could have been tipped but were not named as actual tippees.
- This counterfactual sample is used to test which potential recipients are more likely to receive inside information.

Interpretation. LNPRD is important because actual tipper-tippee links alone cannot reveal which possible relationships were not used. The counterfactual network helps separate “who gets tipped” from “who exists in the tipper’s social circle.”

2.4 Corporate events and returns

- The data include 465 corporate events.
- M&A events account for 239 events, or about 51% of the sample.
- Earnings announcements account for 123 events, or about 26%.

- Other events include clinical-trial and regulatory news, sales of securities, operational announcements, and miscellaneous events.
- The information-to-event return is the return from the date the original source receives the information through the public announcement date.
- Across all events, the mean information-to-event return is about 34.9% over an average of 21.3 trading days.

Interpretation. The events are not ordinary news events. They are high-impact corporate events that create large private trading opportunities.

2.5 People and trading variables

- The sample includes 622 people involved in insider trading networks.
- Some people only pass tips, some only receive tips, and some both give and receive tips.
- About 82.5% of people for whom trading data are available trade stock; about 38.8% trade options.
- The average trader earns very high returns. The person-level mean return is 63.4%, and the median return is 26.4%.
- Occupations include executives, corporate managers, lower-level employees, lawyers, accountants, consultants, buy-side managers, buy-side analysts, small business owners, and specialized occupations.

Interpretation. Inside trading is not only a story about hedge funds. The source network includes corporate insiders and ordinary personal relationships, while the final traders often include buy-side professionals.

2.6 Social relationships and network variables

- A tipper-tippee link is a directed edge from the person who passes information to the person who receives it.
- Relationship types include family, business, friendship, no listed social relation, and more detailed subcategories.
- Key network variables include in-strength, out-strength, in-degree, out-degree, network size, tip-chain distance, density, diameter, and clustering.
- Tip-chain distance is the number of links from the original information source.

Interpretation. The paper studies both local link characteristics, such as family or friendship, and global network position, such as how central a trader is within the overall insider network.

3 Empirical specifications and interpretation

3.1 Tip likelihood using counterfactual potential recipients

The counterfactual tip-recipient analysis can be summarized as a logit model:

$$\Pr(Tip_{ij} = 1) = \Lambda(\alpha + \beta_1 SameGender_{ij} + \beta_2 AgeDiff_{ij} + \beta_3 Family_{ij} + \gamma' RelationType_{ij} + \mu_i), \quad (1)$$

where i indexes the tipper, j indexes a potential tippee, and μ_i is a tipper fixed effect in some specifications.

Interpretation. A positive coefficient means that a potential recipient is more likely to receive a tip relative to other people in the same tipper's social network.

3.2 Daily price-impact regressions

The paper estimates event-firm daily return regressions of the form

$$R_{e,t} = \alpha_e + \lambda_t + \beta InsiderTrading_{e,t} + \Gamma' Controls_{e,t} + \varepsilon_{e,t}, \quad (2)$$

where e is the event firm and t is a pre-announcement trading day.

Alternative trading measures include

$$InsiderTrading_{e,t} \in \{1\{Trade_{e,t} > 0\}, \ln(1 + \#Traders_{e,t}), \ln(1 + \#Shares_{e,t})\}. \quad (3)$$

Controls include market, SMB, HML, volume, event fixed effects, and event-time fixed effects.

Interpretation. The key test is whether illegal insider trading on a given day pushes the stock price in the direction of the eventual public-news return. A positive coefficient means insider trading improves price efficiency before the announcement.

3.3 Individual trading-gain regressions

For people who traded, the paper estimates

$$Y_i = \alpha + \beta_1 Centrality_i + \beta_2 NetworkSize_i + \beta_3 TipDistance_i + \beta_4 \ln(AmountInvested_i) + \beta_5 Options_i + u_i, \quad (4)$$

where

$$Y_i \in \{\ln(1 + Return_i), \ln(1 + Profit_i)\}. \quad (5)$$

Interpretation. If central traders earn higher returns, the result is not merely that they trade more. It means network position gives access to more valuable information.

3.4 Tip-chain analysis

The paper divides each tip into its order in the chain:

$$Order \in \{1, 2, 3, \geq 4\}. \quad (6)$$

The first order is the link from the original source to the first recipient. Higher orders are farther away from the original source.

Interpretation. This design tracks how information changes as it diffuses. Close family and friendship ties dominate early, while business ties and professional traders become more important later.

3.5 Network structure

For each insider network, the paper computes:

- **Density:** realized links divided by all possible links.
- **Diameter:** the longest shortest path between two nodes.
- **Average cluster:** the average fraction of a node's neighbors that are also linked to each other.

Interpretation. Large networks become sparse and branch-like. This is consistent with secrecy: insiders do not all know one another, and information travels through trusted chains.

4 Identification and empirical design

4.1 Why direct communication data matter

- The paper does not infer social interaction from geography, school ties, or correlated trades alone.
- It observes actual legal allegations of information transmission between named people.
- This allows the paper to distinguish direct information sharing from similarity in background or investment style.

Interpretation. The main empirical advantage is direct observation of communication, not a causal shock. The paper’s credibility comes from the quality of the hand-collected legal data.

4.2 Why the counterfactual network matters

- Actual tipper-tippee links show who received tips, but not who could have received tips.
- LNPRD identifies family members and associates who were possible recipients.
- This permits logit tests of the traits associated with receiving a tip.

Interpretation. This is a stronger test than simply reporting that actual tippees are friends or relatives. It asks whether those ties are selected relative to other possible social ties.

4.3 Why the price-impact test is useful

- Insider trading is observed before public announcements.
- Event-day returns reveal the direction of the fundamental news.
- If insider trading moves pre-announcement prices in that direction, it suggests private information is incorporated into prices through the network.

Interpretation. The price-impact result connects private person-to-person communication to market-level price efficiency.

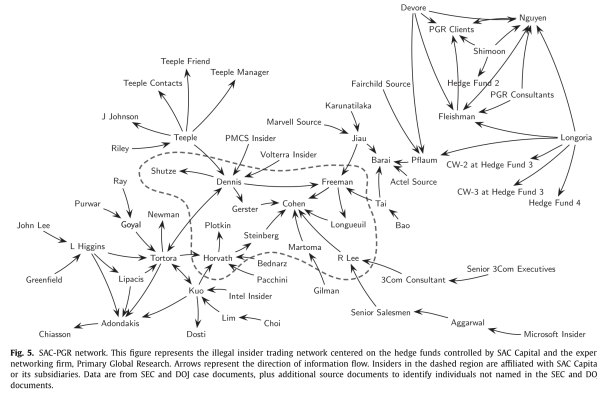
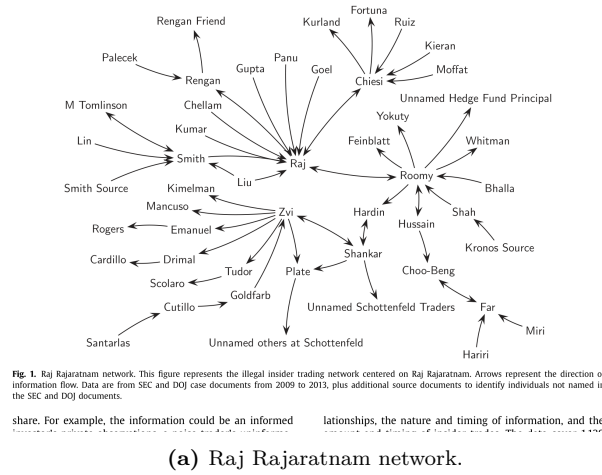
4.4 Main limitations

- The sample contains prosecuted or investigated illegal insider trading, not all insider trading.
- Enforcement incentives likely over-represent large, profitable, repeated, and widespread schemes.
- Legal documents may omit unnamed traders, failed tips, or relationships that prosecutors did not need to prove.
- The setting is illegal tips, so the results may not generalize perfectly to legal information sharing.

Interpretation. The paper is best read as direct evidence on an important subset of information networks, not as a random sample of all investor communication.

5 Tables and figures

5.1 Figure 1 and Figure 5: two large insider-trading networks



Read:

- Figure 1 shows the Raj Rajaratnam-centered network, including many sources and paths that eventually reach Raj, Roomy Khan, Zvi Goffer, and others.
- Figure 5 shows the largest network in the sample, centered on SAC Capital and Primary Global Research.
- Arrows show the direction of information flow.
- The networks are not simple stars. They contain branches, chains, and multiple local clusters.

Economic message. Valuable information moves through a web of personal and professional ties. The shape of the network matters because each link affects speed, secrecy, and trading gains.

5.2 Table 1 and Table 2: events, returns, and event firms

Table 1

Corporate events and returns.

This table presents statistics for 465 corporate events in which the SEC or DOJ identified illegal insider trading, over the years 1996–2013. Panel A reports the number of occurrences of an event type. Panel B reports the average raw stock return from the date that the original tipper receives the information through the announcement date of the event. Panel C reports the average number of trading days over the same time period. Panel D reports the average raw stock return on the announcement date of the event. Negative and Positive columns indicate the expected effect on stock prices, based on the trading behavior of tippees. All includes 18 events that cannot be classified as positive or negative. Stock returns in the All column are calculated as long positive events and short negative events.

Event type	All		Outcome			
			Positive		Negative	
Panel A: Frequency of events						
Clinical trial/drug regulation	37		24		13	
Earnings	123		66		54	
M&A	239		234		5	
Operations	13		8		3	
Sale of securities	35		1		34	
Other	18		2		3	
Total	465		335		112	
	All		Positive		Negative	
	Mean	Median	Mean	Median	Mean	Median
Panel B: Information to event returns (%)						
Clinical trial/drug regulation	79.7	37.1	101.2	36.8	−38.6	−41.7
Earnings	13.5	10.2	14.7	10.4	−12.2	−10.2
M&A	43.1	35.5	43.7	35.7	−20.3	−17.9
Operations	24.9	18.9	22.9	22.2	−29.7	−12.9
Sale of securities	12.8	12.4			−12.8	−12.4
Other	8.9	0.0	−4.8	−4.8	−28.9	−28.9
Total	34.9	23.3	42.4	29.6	−16.8	−12.7
Panel C: Information to event time (trading days)						
Clinical trial/drug regulation	9.2	8.5	11.2	10.0	5.3	3.0
Earnings	11.3	7.0	12.3	8.5	10.2	5.3
M&A	30.5	16.0	31.1	16.5	7.8	4.0
Operations	6.1	3.0	7.9	4.0	2.0	2.0
Sale of securities	10.9	8.0	5.0	5.0	11.1	8.0
Other	44.3	4.0	3.0	3.0	101.1	15.3
Total	21.3	10.0	25.0	11.0	12.2	7.0
Panel D: Announcement day returns (%)						
Clinical trial/drug regulation	50.0	32.3	52.4	9.3	−45.4	−43.3
Earnings	9.2	7.0	7.9	6.0	−10.7	−11.4
M&A	21.9	11.0	22.3	11.3	−6.7	−6.9
Operations	16.9	7.1	12.1	3.2	−28.0	−10.0
Sale of securities	0.2	−0.8			−0.2	0.8
Other	4.4	0.0	−2.7	−2.7	−11.2	−11.2
Total	18.9	7.1	21.8	8.5	−12.0	−5.8

turns of 34.9% over 21.3 trading days. Because the returns to the total returns (Panel B) reveals that announcement

(a) Corporate events and returns.

Table 2

Event firm summary statistics.
This table presents summary statistics of the firms whose stocks are illegally traded based on inside information, using data from 461 corporate events over the years 1996–2013 in which the SEC or DOJ identified illegal insider trading. Total dollars invested by tippees is the total dollar amount of a company's stock purchased or sold across all trades by all inside traders in the data. Daily invested/dollar volume is the average daily dollars invested by tippees divided by daily dollar volume.

	Mean	S.D.	Min	Percentiles			Max	Observations
				25th	50th	75th		
Market equity (billions)	10.09	37.38	0.01	0.30	1.01	3.56	422.04	391
Employees (thousands)	11.71	41.61	0.00	0.37	1.39	6.55	398.46	387
Market V (b)	2.94	2.41	0.35	1.31	1.87	5.68	36.24	391
Daily trading volume (millions)	1.18	8.85	0.00	0.21	0.68	1.85	71.98	391
Daily dollar trading volume (millions)	194.32	372.01	0.02	2.64	11.08	47.19	345.96	391
Total dollars invested by tippees (millions)	4.06	12.98	0.01	0.08	0.37	1.43	132.04	269
Daily invested/dollar volume (%)	4.66	12.48	0.00	0.09	0.63	3.71	95.44	245

(b) Event-firm summary statistics.

Read:

- Table 1 shows that M&A is the dominant event type: 239 of 465 events.
- Average information-to-event returns are very large: 34.9% overall, 43.1% for M&A, and 79.7% for clinical-trial or drug-regulation news.
- The average holding period is short: 21.3 trading days overall and 30.5 days for M&A.
- Table 2 shows that event firms are often large, with mean market equity of \$10.09 billion and median market equity of \$1.01 billion.
- Total dollars invested by tippees are large on average, but the distribution is highly skewed.

Economic message. These are high-value, short-horizon information events. The economic stakes are large enough to explain why traders take legal risk to pass and trade on tips.

5.3 Table 3 and Table 4: who are inside traders?

Table 3

Summary statistics of people involved in insider trading.

This table presents summary statistics for the 622 people in the sample of insider trading from 1996 to 2013. Data are from SEC and DOJ case documents, plus additional sources detailed in the paper. Median house value (median house size) is the median estimated value (square footage) over all properties owned by a person that are also listed as a person's residence in the LexisNexis data. Home values and square footage are as of September 2014 from Zillow. Tips given is the number of insider trading tips given to others. Tips received are the number of insider tips received. Total invested is the aggregated dollar amount of all trading positions in absolute value. Thus, this includes the size of short positions. Total gains are the aggregated dollar amount received by a person across all events and trades. Average return is the average of a person's trades, based on actual buy and sell dates from the SEC and DOJ documents.

	Mean	S.D.	Min	Percentiles			Max	Observations
				25th	50th	75th		
Panel A: Demographics								
Age	44.1	11.5	19.0	35.8	42.7	51.5	80.0	454
Female (%)	9.8	29.8	0.0	0.0	0.0	0.0	100.0	498
Median house value (\$1000s)	1114.3	1846.5	49.6	390.0	656.3	1170.2	25600.0	365
Median house size (1000 ft. ²)	30.0	21.3	7.5	18.6	26.5	35.4	316.4	351
Panel B: Information sharing								
Tips given	1.5	3.2	0.0	0.0	1.0	1.0	29.0	622
Tips received	1.5	2.5	0.0	0.0	1.0	1.0	24.0	622
Panel C: Trading								
Ever traded stock	82.5	38.1	0.0	100.0	100.0	100.0	100.0	399
Ever traded options	38.8	48.8	0.0	0.0	0.0	100.0	100.0	399
Total invested (\$1000s)	4288.0	25334.9	4.4	74.5	226.0	1170.2	375317.3	255
Average invested per tip (\$1000s)	1696.3	6088.6	4.4	65.0	200.0	701.4	72427.5	255
Total invested/median house value (%)	581.6	3609.5	0.6	13.2	38.9	140.3	44183.1	159
Total gains (\$1000s)	2331.6	13207.1	0.9	34.2	136.0	606.0	139500.0	399
Average gains per tip (\$1000s)	1289.9	10036.0	0.1	20.6	72.4	285.0	139000.0	399
Average return (%)	63.4	231.8	0.0	14.0	26.4	46.5	3347.3	255

Table 4

This table presents averages of age, gender, tipping activity, investment, and returns by occupation for the 622 people in the sample of insider trading from 1996 to 2013. Data are from SEC and DOJ case documents, plus additional sources detailed in the paper. Table 3 defines the variables. Internet Appendix Table 3 presents a breakdown of occupations into narrower categories.

Occupation	Count	Percent	Age	Female	House value (\$1,000s)	Tips given	Tips received	Median invested per tip (\$1,000s)	Median gains per tip (\$1,000s)	Median return
Top executive	107	17.3	50.9	6.1	1152	15	0.5	377	2903	33.7
Corporate manager	55	8.9	41.3	5.6	624	16	0.6	1973	286	71.8
Lower-level employee	59	9.5	41.5	25.5	621	17	1.5	558	149	47.5
Sell side/lawyer/accountants	61	9.9	41.7	11.1	1164	3.0	1.9	3,796	255	58.8
Buy side: manager	60	9.7	42.6	6.4	1006	14	2.6	5,972	5,788	37.0
Buy side: analyst/trader	65	10.5	35.5	5.2	1061	2.7	3.1	2,051	442	117.7
Small business owner	39	6.3	47.4	5.3	678	0.9	2.3	204	196	62.0
Specialized occupation	38	6.1	52.0	2.8	729	1.2	1.6	599	234	32.8
Unknown	135	21.8	41.5	20.4	613	0.5	1.1	584	355	77.0

including 13 accountants, 24 attorneys, four investment bankers, and three sell-side analysts. Consistent with

cause identifying the specific timing of real estate purchases is difficult in the NBERD, for each insider trader

(a) Person-level summary statistics.

(b) Summary statistics by occupation.

Read:

- The average inside trader is 44.1 years old; only 9.8% are female.
- The median person receives one tip and gives one tip, but the maximum values are much larger.
- Median amount invested per tip is about \$200,000, while the mean is much larger because of large professional trades.
- Top executives are the largest occupation category, but buy-side managers and analysts are also important.
- Buy-side analysts have high median returns, while buy-side managers have high dollar gains.

Economic message. Illegal inside information begins with people close to the firm but eventually reaches sophisticated, well-capitalized traders.

5.4 Table 5 and Table 6: social relationships and geography

Table 5

Tippees and tipper social relationships.

This table reports the frequency of different types of social relationships among the 461 pairs of people in the insider trading data. When one person tips insider information to another person, they are considered a pair. Pairs can have more than one type of social relationship, so the sum of relationship types is greater than 445. The type of relationship is defined based on the text in SEC and DOJ case documents. Business associates are people who work together or know each other through business relationships where neither person has a supervisory role over the other. Boss refers to business relationships where one person is subordinate to another. Client is a relationship where one person is business client of the other. Data are from SEC and DOJ illegal insider trading case documents filed between 2009 and 2013.

Type of relationship	Count	Fraction of all pairs	Fraction of relationship type
Family	104	22.6	
Dating/engaged	7	1.5	6.7
Married	15	3.3	14.4
Parent-child	20	4.3	19.2
Siblings	25	5.4	24.0
In-laws	12	2.6	11.5
Other	9	2.0	8.7
Unspecified	16	3.5	15.4
Business	160	34.7	
Business associates	87	18.9	54.4
Boss	41	8.9	25.6
Client	32	6.9	20.0
Friends	162	35.1	
Acquaintances	3	0.7	1.9
Friends	115	24.9	71.0
Close friends	44	9.5	27.2
No social relation listed	98	21.3	

Geographic distance by relationship.

Geographic distance is measured in miles using the great circle distance between the cities of residence of the people in a pair. Residences are from the SEC and DOJ case documents. Acquaintances and unknown family relations are omitted because they have very few observations. Table 5 provides a description of the sample.

	Mean	S.D.	Min	Percentiles			Max	Observations
				25th	50th	75th		
All	581.1	1190.8	0	0.0	26.2	739.3	8065.9	229
Family	466.3	1080.1	0	0.0	14.3	322.5	5350.4	59
Married/dating	47.3	183.4	0	0.0	0.0	0.0	710.2	15
Parent-child	326.8	603.6	0	14.3	26.2	160.1	1986.5	13
Siblings	783.3	1490.8	0	6.2	28.0	1063.6	5350.4	14
In-laws	1110.5	1751.5	0	9.6	216.5	1626.3	5350.4	9
Other	88.0	128.3	0	8.1	11.5	237.3	307.6	7
Business ties	327.5	709.5	0	3.5	18.9	219.9	4452.0	98
Associates	334.5	751.6	0	0.0	16.8	236.8	4452.0	57
Boss	198.4	395.1	0	3.9	24.8	53.4	1235.4	22
Client	455.9	857.2	0	10.7	19.0	625.6	2666.9	19
Friendship	715.1	1352.4	0	4.0	28.4	1045.8	8065.9	106
Friends	708.2	1247.7	0	4.5	32.1	1077.6	6622.1	72
Close friends	694.6	1581.8	0	0.0	26.2	835.3	8065.9	32
No social relation listed	962.2	1680.0	0	15.0	80.9	1117.4	5358.7	14

alysts have the highest median return of 117.7%. Buy-side

(a) Tipper-tippee social relationships.

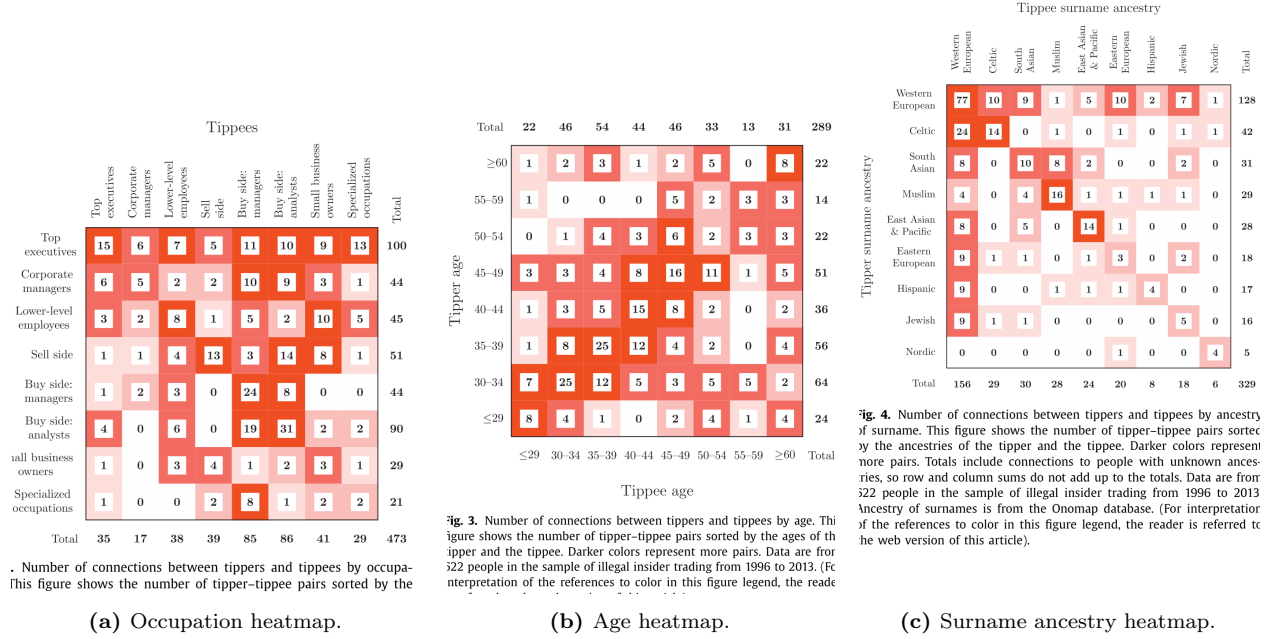
(b) Geographic distance by relationship.

Read:

- Table 5 shows that 23% of observed links are family, 35% are business-related, and 35% are friendships. Categories can overlap.
- Siblings and parent-child links are the most common family ties.
- Business associates are the largest business-tie category.
- Table 6 shows that the median distance between all tipper-tippee pairs is 26.2 miles.
- Median distance is only 14.3 miles for family relations, 18.9 miles for business ties, and 28.4 miles for friendships.

Economic message. Information sharing depends on trust and proximity. Close ties make illegal information transmission more feasible because they reduce fear of detection and betrayal.

5.5 Figures 2–4: shared attributes and homophily



Read:

- Figure 2 shows strong occupation patterns: executives often tip executives and specialized occupations, while buy-side traders become important recipients.
- Figure 3 shows a strong diagonal pattern: people tend to tip others close in age.
- Figure 4 shows that tippers and tippees often share surname ancestry, even beyond family links.

Economic message. The network is socially local. Information does not spread randomly across all investors; it travels through clusters of similar people with shared backgrounds.

5.6 Table 7 and Table 8: who gets tipped and how tips move through chains

Table 7

The likelihood of receiving a tip using potential tippees.

This table presents logit regression coefficients where the dependent variable equals one if a person receives an insider trading tip and zero otherwise. Observations include a tipper's family members and person associates. Family members and person associates are identified using the LexisNexis database. Person associates are non-family members who have a connection to the tipper. Columns 3 and 4 include tipper fixed effects. In columns 2 and 4, the omitted benchmark category is person associates, which means the coefficients on the family members are relative to the likelihood that a person associate receives a tip. Other are family members other than immediate family and in-laws, such as uncles and grandparents. Standard errors are clustered at the tipper level. Table 5 provides a description of the sample. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Likelihood of receiving a tip			
	(1)	(2)	(3)	(4)
Same gender	1.447*** (-0.001)	1.351*** (-0.001)	1.223*** (-0.001)	1.155*** (-0.001)
Age difference	-0.049*** (-0.001)	-0.063*** (-0.001)	-0.049*** (-0.001)	-0.062*** (-0.001)
Family member	-1.097*** (-0.001)		-0.618*** (-0.001)	
Omitted baseline category: Person associates				
Husband		1.651*** (0.007)		1.862*** (-0.001)
Wife		-1.862*** (0.003)		-1.513*** (0.017)
Brother		-0.435 (0.128)		-0.111 (0.704)
Sister		-2.820*** (0.007)		-2.407*** (0.018)
Father		-0.024 (0.957)		0.652 (0.157)
Mother		-1.057 (0.171)		-0.387 (0.613)
Son		0.468 (0.433)		0.698 (0.183)
Daughter		-0.300 (0.781)		-0.223 (0.831)
Other		0.644 (0.205)		0.987** (0.042)
In-law		-1.588*** (-0.001)		-1.070*** (0.003)
Unknown		-3.907*** (-0.001)		-3.569*** (-0.001)
Tipper fixed effects	No	No	Yes	Yes
Pseudo R ²	0.187	0.252	0.099	0.150
Observations	2,604	2,604	2,604	2,604

(a) Likelihood of receiving a tip.

Table 8

Characteristics by order in tip chain.

This table presents the characteristics of tipplers and tippees for each link in a tip chain. The order in a tip chain is the distance from the original source of the inside information, where distance is measured as information links between people. Tables 3 and 5 provide descriptions of the sample.

	Order in tip chain			
	1	2	3	≥ 4
<i>Panel A: Tipper occupation</i>				
Top executive	34.0	6.6	0.8	0.0
Corporate manager	13.7	8.6	3.1	0.0
Lower-level employee	16.1	5.2	11.5	3.6
Sell side/lawyer/accountants	26.3	6.9	29.8	7.3
Buy side: manager	1.0	15.2	17.6	12.7
Buy side: analyst/trader	1.9	27.9	27.5	70.9
Small business owner	0.7	6.9	7.6	1.8
Specialized occupation	1.4	13.4	1.5	0.0
Unknown	4.8	9.3	0.8	3.6
<i>Panel B: Tipper demographics</i>				
Female	11.5	14.8	4.5	10.5
Age	42.8	41.4	35.7	34.4
Median house value (\$1,000s)	811.7	840.1	758.5	1,072.0
<i>Panel C: Connections</i>				
Family connection	24.6	15.5	28.4	11.9
Friendship connection	42.4	35.2	36.6	18.6
Business connection	28.4	47.0	34.3	66.1
No connection	20.9	19.7	15.7	11.9
Geographic distance – median (miles)	15.8	40.2	46.9	0.0
Same house value quintile	49.6	36.2	15.2	20.0
Same surname ancestry	51.9	33.8	42.1	42.5
<i>Panel D: Trading</i>				
Amount invested – median (\$1,000s)	200.4	250.1	280.1	492.7
Gross profit – median (\$1,000s)	17.6	36.3	39.5	86.0
Tip return – average (%)	46.0	43.5	29.2	23.0
Tip return – median (%)	25.2	27.9	28.2	18.8
Use shares dummy (%)	50.8	56.2	77.1	76.4
Use options dummy (%)	27.0	23.4	16.8	21.8
<i>Panel E: Timing</i>				
Time lapse from information to tip (days)	12.1	9.2	5.0	0.4
Tip passed on same day as received (%)	46.5	62.7	49.5	92.1
Holding period – average (days)	13.9	16.8	11.3	9.1
Holding period – median (days)	5.2	7.0	4.0	5.0

(b) Characteristics by order in tip chain.

Read:

- Table 7 shows that same-gender potential tippees are more likely to receive tips.
- Age difference is negatively related to receiving a tip.
- Family members are less likely to be tipped than non-family associates on average, but husbands and some other relatives are more likely to be tipped than associates.
- Table 8 shows that original sources are most often executives and corporate insiders.
- By the fourth and later links, buy-side analysts and traders account for 70.9% of tippees.
- Business connections rise over the chain, while family and friendship connections become less dominant.

Economic message. The first transmission depends on trust and close ties. Later transmission moves toward professional traders who can monetize the information more aggressively.

5.7 Table 9: network size and structure

Table 9

Tipper and tippee characteristics by size of insider network.

This table reports characteristics of people in information networks of increasing size. Size of network refers to the number of people in an insider network (where every member can be reached by every other member through at least one path). Density is the proportion of all possible connections that actually exist. The diameter of a network is the longest of all shortest paths between any two members. Average cluster is the average node's clustering, where clustering is the fraction of a node's links that are also linked to each other. Tables 3 and 5 provide descriptions of the sample.

	Size of network			
	1	2	3-5	≥ 6
<i>Panel A: Network characteristics</i>				
Density		1.0	0.6	0.2
Diameter		1.0	2.3	4.4
Average cluster		2.0	0.2	0.1
<i>Panel B: Personal characteristics</i>				
Female tipper (%)	3.4	22.7	11.7	7.2
Female tippee (%)		8.6	10.6	8.5
Tipper age	47.8	45.4	43.6	42.1
Tippee age		44.9	44.5	42.3
Tippee house value – mean (\$1,000s)		856.8	893.2	1311.6
Tippee house value – median (\$1,000s)		594.8	476.2	768.0
Tipper house value – mean (\$1,000s)	849.6	884.2	796.2	1548.4
Tipper house value – median (\$1,000s)	572.7	560.7	620.6	960.8
<i>Panel C: Social connections</i>				
Family connections (%)		43.2	28.2	23.8
Business connections (%)		20.5	34.0	36.4
Friendship connections (%)		36.4	37.8	39.7
Geographic distance – mean (miles)		523.8	326.0	583.6
Geographic distance – median (miles)		7.2	33.6	36.6
<i>Panel D: Trading</i>				
Amount invested – mean (\$1,000s)	725.3	1054.0	1227.7	2457.3
Amount invested – median (\$1,000s)	131.6	294.8	154.9	303.1
Gross profit – mean (\$1,000s)	2666.1	1248.1	255.7	526.6
Gross profit – median (\$1,000s)	67.4	154.3	41.7	216.1
Tip return – mean (%)	137.7	82.0	51.0	37.3
Tip return – median (%)	32.4	34.8	34.2	28.0
<i>Panel E: Timing</i>				
Time lapse from information to tip (days)		11.3	15.4	9.3

clustering coefficient is the fraction of a node's links that are also linked to each other. The average clustering coefficient

Figure 7: Tipper and tippee characteristics by size of insider network.

Read:

- Larger networks are less dense and have lower clustering.
- Network diameter increases with size: large networks are long and branch-like.
- Larger networks include more business relationships and larger dollar investments.
- Tip returns tend to be lower in larger networks even though dollar profits can be large.

Economic message. The structure resembles a criminal network: information is passed through limited trusted paths rather than broadcast to everyone.

5.8 Table 10 and Table 11: prices and individual gains

Table 10

Insider trading and daily returns.

This table presents fixed effects regressions on daily stock returns for the event firms in the sample. Observations are from 120 trading days before the public announcement of the event to one day before the announcement. All regressions include event fixed effects, which controls for all characteristics of the event and of the firm that are time invariant over the 120 trading days before the announcement. All regressions also include $\ln(1+\text{daily volume})$ and daily returns for the market, SMB, and HML factors. Event-time fixed effects are dummy variables for days in the $(-120, -1)$ period. Insider trading days only include observations in which an insider traded on a particular day. Insider trading dummy is a dummy variable that takes the value of one if an insider traded on a particular day. $\ln(1+\# \text{ of insider traders})$ is the log of one plus the number of unique insider traders that trade on a particular day. $\ln(1+\# \text{ of shares traded by insider traders})$ is the log of one plus the total number of shares traded by insiders on a particular day. Table 1 describes the sample. p -values from standard errors clustered at the event level are presented in parentheses. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	Dependent variable: daily stock return (%)					
	(1)	(2)	(3)	(4)	(5)	(6)
Insider trading dummy	0.204*** (0.006)					
$\ln(1+\# \text{ of insider traders})$		0.270*** (0.001)	0.197** (0.024)			
$\ln(1+\# \text{ of shares traded by insider traders})$				0.109*** (-0.001)	0.087*** (0.007)	0.288** (0.049)
Control variables:						
Volume, MKTRET, SMB, and HML	Yes	Yes	Yes	Yes	Yes	Yes
Event fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Event-time fixed effects	No	No	Yes	No	Yes	Yes
Insider trading days only	No	No	No	No	No	Yes
Observations	48,445	48,445	48,445	48,445	48,445	2,338
Adjusted R^2	0.154	0.155	0.155	0.155	0.155	0.135

indicate that tips from family members are more reliable, without trading. The dependent variable in columns 1-3 is $\ln(\text{returns})$, where returns are in percentages. 1-win which leads to more aggressive trading. In contrast, tips

(a) Insider trading and daily stock returns.

Table 11

Insider characteristics and individual trading gains.

This table presents OLS regressions on daily stock returns for the event firms in the sample. Observations are for all insider traders that traded securities for whom data are available. The dependent variable in columns 1-3 is the log of one plus the insider trader's total return over all trades, calculated as total profit divided by total invested. The dependent variable in columns 4-6 is the log of the insider trader's total profit over all trades. Network instrength is the number of tips received by a trader in the entire sample. Network outstrength is the number of tips given by a trader in the entire sample. Network indegree is the number of different tippees that gave tips to an insider. Network outdegree is the number of tippees to whom a trader gave tips. Network size is the number of insider traders in a trader's network of insiders. $\ln(\text{Amount invested})$ is the total dollar amount of the insider trader's investments. Tip chain distance is the number of links from the original source. Uses options is a dummy variable that indicates if a trader ever traded options on inside information. Table 1 describes the sample. p -values from robust standard errors are presented in parentheses. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *.

	$\ln(\text{Return})$			$\ln(\text{Profit})$		
	(1)	(2)	(3)	(4)	(5)	(6)
Network outstrength	-0.025 (0.225)		0.010 (0.702)	-0.030 (0.489)		-0.022 (0.753)
Network instrength	0.085*** (-0.001)		0.085*** (-0.001)	0.217*** (-0.001)		0.215*** (-0.001)
Network outdegree		-0.086 (0.153)	-0.130 (0.129)		-0.009 (0.918)	-0.029 (0.848)
Network indegree		0.128 (0.078)	-0.023 (0.789)		0.365** (0.022)	0.005 (0.976)
Network size	0.012** (0.029)	0.012** (0.042)	0.014** (0.023)	0.042*** (-0.001)	0.041*** (-0.001)	0.042*** (-0.001)
Tip chain distance	-0.007 (0.906)	0.015 (0.809)	-0.011 (0.856)	-0.231*** (0.009)	-0.191** (0.036)	-0.232*** (0.010)
$\ln(\text{Amount invested})$	-0.309*** (-0.001)	-0.278*** (-0.001)	-0.311*** (-0.001)			
Use options	1.545*** (-0.001)	1.665*** (-0.001)	1.545*** (-0.001)	1.237*** (-0.001)	1.239*** (-0.001)	1.237*** (-0.001)
Constant	7.680*** (-0.001)	6.658*** (-0.001)	7.147*** (-0.001)	10.988*** (-0.001)	10.988*** (-0.001)	10.993*** (-0.001)
Observations	249	249	249	399	399	399
Adjusted R^2	0.311	0.281	0.311	0.247	0.189	0.243

(b) Insider characteristics and individual gains.

Read:

- Table 10 shows that insider-trading days are associated with positive event-stock returns before public announcements.
- The insider-trading dummy coefficient is 0.204 percentage points.
- More inside traders and more shares traded are also associated with larger daily stock returns.
- Table 11 shows that network in-strength is positively related to both logged return and logged profit.
- Network size is also positively related to returns and profits.
- Tip-chain distance is negatively related to profits: information becomes less valuable as it travels farther from the source.

Economic message. Social networks matter for both market efficiency and private gains. Information reaches prices gradually, and better-positioned traders earn more.

6 Short summary

- The paper studies direct evidence of social information transmission using illegal insider trading cases.
- The sample contains 183 networks, 622 people, 1,139 tips, 465 corporate events, and \$928 million in illegal profits.
- Inside information is shared mostly through strong ties: family, friends, business associates, and geographically close relationships.
- The average information-to-event return is about 35% over 21 trading days.
- M&A and clinical/regulatory announcements are especially valuable information events.
- Information usually begins with corporate insiders and moves toward buy-side professionals.
- Same gender, similar age, geographic proximity, school ties, and shared surname ancestry all help characterize tipper-tippee links.
- Lower-status people often tip higher-status people, suggesting social hierarchy and favor-seeking.
- Large insider networks are sparse and branch-like.
- Insider trading moves prices in the direction of eventual public information.
- More central traders earn larger returns and profits.
- The main limitation is sample selection: the data cover caught and prosecuted insider trading, not all private information sharing.

7 Conclusion

7.1 Core conclusion

Ahern provides direct evidence that financial information spreads through social networks. Illegal insider tips flow through close, trusted relationships and then diffuse toward professional investors who can trade at scale. These social paths affect both stock prices and individual trading profits.

7.2 Economic intuition

Inside information is valuable but dangerous to share. A tipper wants to share it with someone who can be trusted and who may reciprocate or provide social benefits. That pushes information into dense local relationships first. As the tip moves farther from the source, it becomes less secret but reaches traders with more capital and sophistication. Prices incorporate the information gradually along this path.

7.3 Takeaway

The paper shows that investor communication is not an abstract market friction. It is a network process built on family, friendship, geography, status, and professional relationships. Social networks help determine who learns first, who trades most profitably, and how quickly prices become informed.